

Introduction to Data Science — Final Exam Paper 2 (Easy)

Course Code: BCSE2C05 | Semester: II | Max Marks: 100 | Time: 3 Hours Difficulty: ★☆☆☆☆ Easy

SECTION A — Short Answer (5 Marks Each, Do Any 4 of 5 = 20 Marks)

Q1. What is Data Science? List any four applications of Data Science.

Data Science is an interdisciplinary field that uses scientific methods, algorithms, and systems to extract knowledge and insights from structured and unstructured data.

Four Applications:

- 1. **Healthcare** — Predicting disease outbreaks, drug discovery, patient diagnosis.
- 2. **E-Commerce** — Product recommendation systems (Amazon, Flipkart).
- 3. **Finance** — Fraud detection in credit card transactions.
- 4. **Social Media** — Sentiment analysis, targeted advertising (Facebook, Instagram).

Q2. Differentiate between Discrete and Continuous data with examples.

Aspect	Discrete Data	Continuous Data
Definition	Countable, finite values.	Measurable, can take any value within a range.
Values	Whole numbers only.	Decimal/fractional values possible.
Example	Number of students in a class (30, 31, 32)	Height of students (5.5 ft, 5.7 ft, 6.1 ft)
Visualization	Bar chart	Histogram

Q3. Write Python code to create a NumPy array and find its mean, sum, and maximum value.

```
import numpy as np

arr = np.array([10, 20, 30, 40, 50])

print("Mean:", np.mean(arr))      # Output: 30.0
print("Sum:", np.sum(arr))        # Output: 150
print("Max:", np.max(arr))        # Output: 50
print("Min:", np.min(arr))        # Output: 10
print("Std Dev:", np.std(arr))    # Output: 14.14
```

Q4. What is the difference between Mean, Median, and Mode? Give an example.

Measure	Definition	Formula/Method	Example (Data: 2, 3, 3, 5, 7)
Mean	Average of all values	Sum / Count	(2+3+3+5+7)/5 = 4.0
Median	Middle value when sorted	Middle position	Sorted: 2,3, 3 ,5,7 → 3
Mode	Most frequently occurring value	Highest frequency	3 (appears twice)

Q5. What is Matplotlib? Write a code to plot a simple line chart.

Matplotlib is a Python library used for creating static, animated, and interactive visualizations. It is the most widely used plotting library in Python for data science.

```
import matplotlib.pyplot as plt

x = [1, 2, 3, 4, 5]
y = [10, 20, 25, 30, 40]

plt.plot(x, y, marker='o', color='blue', linestyle='-')
plt.title('Simple Line Chart')
plt.xlabel('X-Axis')
plt.ylabel('Y-Axis')
plt.grid(True)
plt.show()
```

SECTION B — Analytical Questions (8 Marks Each, Do Any 5 of 6 = 40 Marks)

Q6. Explain the Data Science Lifecycle with a neat diagram.

Data Science Lifecycle is a systematic approach to solving problems using data:

```
Problem Definition → Data Collection → Data Cleaning → EDA →
Model Building → Evaluation → Deployment
```

Phases:

1. **Problem Definition** — Identify the business question. *Example:* "Which customers will churn?"
 2. **Data Collection** — Gather data from databases, APIs, surveys. *Example:* Customer transaction logs.
 3. **Data Cleaning** — Handle missing values, duplicates, inconsistencies. *Example:* Fill missing ages with median.
 4. **EDA (Exploratory Data Analysis)** — Visualize data using plots, compute statistics. *Example:* Histogram of purchase amounts.
 5. **Model Building** — Train ML algorithms. *Example:* Logistic regression for churn prediction.
 6. **Evaluation** — Test model using accuracy, F1-score. *Example:* 92% accuracy on test data.
 7. **Deployment** — Put model into production. *Example:* REST API for real-time predictions.
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Q7. What is a DataFrame in Pandas? Write code to create a DataFrame and perform basic operations.

A **DataFrame** is a 2D, size-mutable, tabular data structure in Pandas with labelled rows and columns (like a spreadsheet or SQL table).

```
import pandas as pd

# Creating a DataFrame
data = {
    'Name': ['Alice', 'Bob', 'Charlie', 'Diana'],
    'Age': [22, 25, 23, 28],
    'Marks': [85, 90, 78, 92]
}
df = pd.DataFrame(data)

# Basic operations
print(df.head())          # First 5 rows
print(df.shape)           # (4, 3)
print(df.describe())      # Statistical summary
print(df['Age'].mean())    # Mean age = 24.5
print(df.dtypes)          # Data types of each column

# Filtering
top_students = df[df['Marks'] > 80]
print(top_students)

# Sorting
```

```
df_sorted = df.sort_values('Marks', ascending=False)
print(df_sorted)
```

Q8. What are the different types of data in Data Science? Explain with examples.

1. Based on Structure:

Type	Description	Example
Structured	Organized in rows & columns.	SQL databases, CSV files, spreadsheets
Unstructured	No predefined format.	Emails, images, videos, social media posts
Semi-Structured	Partially organized with tags/markers.	JSON, XML, HTML files

2. Based on Attribute Types:

Attribute	Description	Example
Nominal	Categories with no order.	Gender (Male/Female), Color (Red/Blue)
Ordinal	Categories with meaningful order.	Rating (Poor < Average < Good < Excellent)
Binary	Only two values.	Pass/Fail, Yes/No
Numeric (Discrete)	Countable whole numbers.	Number of children: 0, 1, 2, 3
Numeric (Continuous)	Measurable, any value in a range.	Temperature: 36.5°C, Weight: 65.2 kg

Q9. What is Data Preprocessing? Why is it needed?

Data Preprocessing is the process of transforming raw data into a clean, usable format before analysis or model building.

Why it is needed:

- Raw data is often incomplete, inconsistent, noisy, and full of errors.
- ML models cannot work with missing or poorly formatted data.
- Clean data leads to better model accuracy and reliable insights.

Steps of Data Preprocessing:

Step	Description	Example
Data Cleaning	Handle missing values, noise, duplicates.	Fill NaN with mean, remove duplicates.
Data Integration	Combine data from multiple sources.	Merging customer data from CRM and billing.
Data Transformation	Normalize, aggregate, or encode data.	Min-Max normalization, one-hot encoding.
Data Reduction	Reduce volume while preserving information.	PCA, feature selection, sampling.

Q10. Explain the different sources of data with examples.

Source	Description	Example
Time Series Data	Data collected at regular intervals over time.	Daily stock prices, monthly rainfall data.
Transactional Data	Records of business transactions.	E-commerce purchase records, bank transactions.
Biological Data	Data from living organisms and medical studies.	DNA sequences, patient health records.
Spatial Data	Data associated with geographic locations.	GPS coordinates, satellite images, maps.
Social Network Data	Data from social media interactions.	Twitter tweets, Facebook likes, Instagram comments.

Data Evolution: Over time, data formats, volume, and sources evolve — new sources like IoT sensors and wearables generate continuous streams of data requiring real-time processing.

Q11. Write a Python program to handle missing values using Pandas.

```
import pandas as pd
import numpy as np

# Create DataFrame with missing values
data = {
    'Name': ['Alice', 'Bob', 'Charlie', 'Diana', 'Eve'],
    'Age': [22, np.nan, 25, np.nan, 28],
    'Salary': [50000, 60000, np.nan, 45000, np.nan]
}
df = pd.DataFrame(data)

# Check missing values
print("Missing values:\n", df.isnull().sum())

# Method 1: Fill with mean
df['Age'] = df['Age'].fillna(df['Age'].mean())
print("\nAfter filling Age with mean:\n", df)

# Method 2: Fill with median
df['Salary'] = df['Salary'].fillna(df['Salary'].median())
print("\nAfter filling Salary with median:\n", df)

# Method 3: Drop rows with any NaN (on original data)
df_dropped = df.dropna()
print("\nAfter dropping rows with NaN:\n", df_dropped)

# Method 4: Forward fill
df_ffill = df.ffill()
print("\nAfter forward fill:\n", df_ffill)

# Method 5: Backward fill
df_bfill = df.bfill()
print("\nAfter backward fill:\n", df_bfill)
```

SECTION C — Long Answer (20 Marks Each, Do Any 2 of 3 = 40 Marks)

Q12. Explain Big Data and its characteristics. How does Big Data form the foundation of Data Science?

(a) Big Data (10 Marks):

Big Data refers to extremely large and complex datasets that cannot be processed using traditional data management tools.

5 V's of Big Data:

V	Description	Example
Volume	Huge amount of data.	Facebook generates 4+ petabytes daily.
Velocity	Speed of data generation.	Stock market tickers updating every millisecond.
Variety	Different data types.	Text, images, audio, video, sensor data.
Veracity	Data quality and accuracy.	Social media may have fake news/bots.
Value	Usefulness of processed data.	Customer insights leading to better marketing.

Impact on Industries:

1. **Healthcare** — Predictive diagnosis using patient records.
2. **Retail** — Amazon's recommendation engine.
3. **Banking** — Real-time fraud detection.
4. **Transportation** — Uber's surge pricing and route optimization.
5. **Education** — Personalized learning platforms.

(b) Big Data as Foundation of Data Science (10 Marks):

Data Science relies on Big Data because:

- **Data Collection** — Big Data provides the raw material (logs, sensors, social media).
- **Storage** — Technologies like Hadoop and Spark enable storing and processing massive datasets.
- **Analysis** — Statistical and ML techniques extract patterns from large datasets.
- **Visualization** — Tools like Tableau and Matplotlib present findings.
- **Decision Making** — Insights from Big Data drive business strategy.

Case Study — Netflix: Netflix collects viewing data from 200M+ users, analyses it using ML models to recommend content, and even decides which original shows to produce based on data-driven insights.

Q13. Write a Python program to read a CSV file, perform data analysis using Pandas, and create visualizations.

Scenario: A CSV file `students.csv` has columns: `Roll_No`, `Name`, `Age`, `Department`, `CGPA`, `Attendance` .

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Task 1: Read the CSV file
df = pd.read_csv('students.csv')
print("Shape:", df.shape)
print("\nFirst 5 rows:")
print(df.head())
print("\nData Types:")
print(df.dtypes)

# Task 2: Basic statistics
print("\nDescriptive Statistics:")
print(df.describe())
print(f"\nMean CGPA: {df['CGPA'].mean():.2f}")
print(f"Median Age: {df['Age'].median()}")
print(f"Mode of Department: {df['Department'].mode()[0]}")

# Task 3: Handle missing values
print("\nMissing values:")
print(df.isnull().sum())
df['CGPA'] = df['CGPA'].fillna(df['CGPA'].mean())
df['Attendance'] = df['Attendance'].fillna(df['Attendance'].median())

# Task 4: Filter students with CGPA > 8
toppers = df[df['CGPA'] > 8.0]
print(f"\nStudents with CGPA > 8: {len(toppers)}")
print(toppers[['Name', 'CGPA']])

# Task 5: Group by Department
dept_avg = df.groupby('Department')['CGPA'].mean()
print("\nAverage CGPA by Department:")
print(dept_avg)

# Task 6: Histogram of CGPA
```

```
plt.figure(figsize=(8, 5))
plt.hist(df['CGPA'], bins=10, color='steelblue', edgecolor='black')
plt.title('Distribution of CGPA')
plt.xlabel('CGPA')
plt.ylabel('Frequency')
plt.show()

# Task 7: Bar chart of department-wise average CGPA
plt.figure(figsize=(8, 5))
dept_avg.plot(kind='bar', color='coral', edgecolor='black')
plt.title('Average CGPA by Department')
plt.xlabel('Department')
plt.ylabel('Average CGPA')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

# Task 8: Save cleaned data
df.to_csv('students_cleaned.csv', index=False)
print("Cleaned data saved!")
```

Q14. Explain Supervised and Unsupervised Machine Learning. Describe k-NN and K-Means with examples.

(a) Supervised vs Unsupervised Learning (8 Marks):

Aspect	Supervised Learning	Unsupervised Learning
Data	Labelled (input + output known)	Unlabelled (only input)
Goal	Predict output for new inputs	Discover hidden patterns
Types	Classification, Regression	Clustering, Association
Example	Spam detection (spam/not spam)	Customer segmentation
Algorithms	k-NN, Decision Tree, SVM	K-Means, DBSCAN, Apriori

(b) k-Nearest Neighbours (k-NN) (6 Marks):

k-NN is a supervised algorithm that classifies a new point based on the majority class of its k nearest neighbours.

Steps:

1. Choose k (e.g., k = 3).
2. Calculate distance from new point to all training points (Euclidean distance).
3. Pick the k closest points.
4. Assign the majority class.

Example: Classifying a fruit as Apple or Orange based on weight and colour:

- New fruit: Weight=150g → 3 nearest neighbours: 2 Apples, 1 Orange → **Apple**

(c) K-Means Clustering (6 Marks):

K-Means is an unsupervised algorithm that partitions data into K clusters.

Steps:

1. Choose K (number of clusters).
2. Initialize K centroids randomly.
3. Assign each point to the nearest centroid.
4. Recompute centroids as the mean of each cluster.
5. Repeat until convergence.

Example: Given points {1, 2, 8, 9, 10}, K=2, initial centroids C1=1, C2=8:

- **Iteration 1:** Cluster1={1,2}, Cluster2={8,9,10} → C1=1.5, C2=9.0
 - **Iteration 2:** Cluster1={1,2}, Cluster2={8,9,10} → C1=1.5, C2=9.0 (converged ✓)
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— End of Paper 2 (Easy) —